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Issues in the selection of a summary statistic for meta-analysis of clinical trials with binary outcomes

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SUMMARY

Meta-analysis of binary data involves the computation of a weighted average of summary statistics calculated for each trial. The selection of the appropriate summary statistic is a subject of debate due to conflicts in the relative importance of mathematical properties and the ability to intuitively interpret results. This paper explores the process of identifying a summary statistic most likely to be consistent across trials when there is variation in control group event rates. Four summary statistics are considered: odds ratios (OR); risk differences (RD) and risk ratios of beneficial (RR(B)); and harmful outcomes (RR(H)). Each summary statistic corresponds to a different pattern of predicted absolute benefit of treatment with variation in baseline risk, the greatest difference in patterns of prediction being between RR(B) and RR(H). Selection of a summary statistic solely based on identification of the best-fitting model by comparing tests of heterogeneity is problematic, principally due to low numbers of trials. It is proposed that choice of a summary statistic should be guided by both empirical evidence and clinically informed debate as to which model is likely to be closest to the expected pattern of treatment benefit across baseline risks. Empirical investigations comparing the four summary statistics on a sample of 551 systematic reviews provide evidence that the RR and OR models are on average more consistent than RD, there being no difference on average between RR and OR. From a second sample of 114 meta-analyses evidence indicates that for interventions aimed at preventing an undesirable event, greatest absolute benefits are observed in trials with the highest baseline event rates, corresponding to the model of constant RR(H). The appropriate selection for a particular meta-analysis may depend on understanding reasons for variation in control group event rates; in some situations uncertainty about the choice of summary statistic will remain. Copyright © 2002 John Wiley & Sons, Ltd.

KEY WORDS: meta-analysis; odds ratio; risk ratio; risk difference, binary data; randomized controlled trials

1. INTRODUCTION

The starting point of all meta-analyses is the selection of a statistic (effect measure) used to describe the observed treatment effect in each trial, from which the overall meta-analytical summary is calculated. Three alternative measures are generally considered for binary out-

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comes from clinical trials – the odds ratio, the risk difference and the risk ratio – but little guidance is available regarding the choice between them.

The selection of a summary statistic for meta-analyses of binary outcomes is an issue that has been much debated in the literature, selection being argued on the grounds of consistency of effect [1], ease of interpretation [2, 3] and mathematical properties [4, 5]. The issue remains contentious, principally because the estimator with the best mathematical properties (the odds ratio) is the least intuitive. The promotion of a measure often reflects personal preferences – those who are keen to promote the use of research in practice emphasize issues of interpretability of risk ratios and risk differences, those who are keen to ensure mathematical rules are always obeyed emphasize the limitations and inadequacies of the same measures.

In health care, meta-analytical summaries are used for two purposes:

1. to describe the average treatment effect and summarize its statistical significance;
2. to predict likely clinical benefit for future groups of patients.

The first application will not be the focus of this paper. Although the comprehension of the statistics varies, they can all be re-expressed in terms of each other assuming a typical value of baseline risk. Also, it is rare for the statistical significance of a meta-analysis to depend on the summary statistic. Recently Engels *et al.* compared the use of odds ratio and risk difference summary statistics in 125 meta-analyses [6] and did not find a single analysis where the two summary statistics gave different conclusions, such that one indicated significant benefit and the other significant harm.

The second application, assessment of the suitability of each summary statistic for predicting benefits of treatment for future patients, is the main motivation for this paper. A basic concept that underpins the transferability of results of meta-analysis to clinical practice is that the effect of a particular treatment may be constant in different patient groups, despite variations in baseline risk between the groups. Throughout this paper the term *consistency* will be used to describe this property. The challenge is in knowing which of the summary statistics will be the most consistent in a particular situation: is it more likely that the ratio of odds is constant across varying baseline risks, or the ratio of risks, or the difference between risks? If an inappropriate statistic is used in the meta-analysis, the predictions that are made from it will only be correct at the average baseline risk – extrapolation to other baseline risks will be unreliable.

Sections 2 and 3 of the paper briefly outline meta-analytical methods and introduce four relevant examples selected to illustrate particular methodological issues. In Section 4 the summary statistics are conceived as depicting different 'models' of predictions of absolute benefit with baseline risk. The constraints and patterns of each model are explored, and epidemiological support for particular models reviewed.

Walter has argued that the selection of a summary statistic should not be based on mathematical dogma, but on evidence [7, 8]. Where there is variation in the event rates in control groups among the trials included in a review, the summary statistics cannot be equally appropriate summaries of effect (unless there is no treatment effect). One summary will be noted to give a better fit of the observed trial data than the others. Theoretically this could be interpreted as empirical evidence that one particular assumption (constant relative odds or constant relative risk or constant risk difference) is more tenable for predicting future treatment benefit of that particular intervention than the others.

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Unfortunately, for most meta-analyses inadequate numbers of trials are available to sensibly compare goodness-of-fit of the summary statistics. However, comparisons of analyses using the different summary statistics across many meta-analyses could indicate which statistic, on average, is likely to be the most consistent estimator. Engels *et al.* compared the heterogeneity of risk difference analyses with odds ratios analyses in 125 meta-analysis and noted that 14 per cent were significantly heterogeneous ($p<0.1$) when risk difference was used but not when odds ratios were used. Two investigations that are complementary to Engels study are reported in Section 5.

Whilst such aggregate cross-meta-analytical analyses provide general guidelines for selection of a summary statistic, they cannot identify which summary statistic is likely to provide the most appropriate predictions of treatment benefit for a particular meta-analysis. One approach that has been proposed is to select the statistic that yields the lowest heterogeneity statistic. The limitations of this approach are discussed in Section 6.

Engels *et al.* did not examine the risk ratio as a suitable summary statistic [6]. Some statisticians argue against use of a risk ratio due to the asymmetry introduced by switching the coding of the event and non-event in the analysis [9]. For meta-analyses using risk differences or odds ratios the impact of this switch is of no great consequence; the switch simply changes the sign of a risk difference, whilst for odds ratios the new odds ratio is the reciprocal of the original odds ratio. By contrast, switching the outcome can make a substantial difference for risk ratios, affecting the effect size, its significance and observed heterogeneity. In a meta-analysis the effect of this reversal cannot be predicted mathematically.

This paper overcomes the asymmetry issue by considering that there are two risk ratios for every meta-analysis, namely risk ratios of beneficial and harmful outcomes (denoted by RR(B) and RR(H), respectively), and investigates how they differ and whether one appears more consistent in a sample of reviews than the other. To complement the work of Engels *et al.* the performance of the risk ratio as a summary measure for meta-analysis is compared with both the odds ratio and risk difference.

2. STATISTICAL METHODS FOR POOLING RISK RATIOS

Engels summarized methods for pooling risk differences and odds ratios [6]. Three similar simple methods are available for pooling risk ratios [10], two fixed effect approaches (the inverse variance method and a Mantel–Haenszel style estimator described by Greenland and Robins [11]) and one random effects method based on DerSimonian and Laird's approach. Zero event counts in either arm cause computational problems for all methods, but are usually avoided by the addition of 0.5 either to all cells of the problematic tables or to all cells of all trials.

Cochran's heterogeneity statistic, Q, is calculated for all methods using the inverse variance weights w_i as

$$Q = \sum w_i(\hat{\theta}_i - \hat{\theta})^2$$

where θ_i is the log OR, log RR or RD in each trial, and θ the common log OR, log RR or RD estimated from the meta-analysis.

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3. MOTIVATING EXAMPLES

Four examples are presented to illustrate the issues linked to selection of a summary statistic. For each example pooled results are obtained using four summary statistics: the odds ratio; the risk difference; and risk ratios of (a) the beneficial outcome and (b) the harmful outcome. Fixed effect analyses have been undertaken using the Mantel–Haenszel method and random effects analyses were performed according to the DerSimonian and Laird approach. Analyses were undertaken using the `metan` procedure in Stata [12]. Cochran's Q was used to assess heterogeneity.

3.1. Polysaccharide vaccine for preventing Meningitis A [13]

A systematic review pooled results from seven randomized trials with 12-month follow-up comparing groups receiving polysaccharide meningitis A vaccine with either placebo or no vaccine. The seven trials varied in the incidence of meningitis in the control group. In five trials undertaken in endemic areas, annual incidence rates varied between 1.0 and 5.3 per 100 000, whilst two trials coincided with epidemics when annual incidence rates were 35 and 57 per 100 000. Results of analyses are given in Table I, indicating that the vaccine is very effective. As events are rare, analyses of odds ratios and risk ratios of harm are very similar, but there are substantial increases in heterogeneity associated with the risk difference analysis and switching events for risk ratios.

3.2. Eradication of Helicobacter pylori in non-ulcer dyspepsia [14]

Helicobacter pylori is a bacterium that inhabits the stomach and has been considered to have a possible causal role in the development of non-ulcer dyspepsia. Meta-analyses of the five relevant trials of eradication of *H. pylori* in non-ulcer dyspeptics published before 2000 are given in Table I, this being a rare example where the significance of the small beneficial effect depends on the choice of summary statistic. Two systematic reviews have recently been published using data from these trials. The first summarized risk ratios of remaining dyspeptic 12 months after eradication, and concluded significant benefit [15]. The second meta-analysis (published in a different journal) used odds ratios as the outcome measure and the authors concluded that there is no significant benefit of eradicating *Helicobacter pylori* in these patients [16]. Although there were differences between the reviews other than the choice of summary statistic, if the authors of the second review had summarized their data using risk ratios of remaining dyspeptic thus would have reached the alternative conclusion of significant benefit. (More recently the Cochrane review has been updated to include trials published in 2000 [17], all analyses now being statistically significant, but for the purpose of this paper we will consider meta-analyses of the five trials published before 2000.)

3.3. Single-dose aspirin 600–650 mg for treating acute pain [18]

The third example summarizes results from 64 trials comparing single-dose aspirin with placebo for the treatment of acute pain related to surgical procedures. The outcome used in the trial is the proportion obtaining 50 per cent pain relief. Where this is not reported directly values were imputed based on transformations of alternative pain scale outcomes. All

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Table I. Results of meta-analyses of constant OR, RD, RR(B) and RR(H) for the four case-studies introduced in Section 3.

Measure	Meningitis vaccination [13]			Eradication of *H. Pylori* [14]			Single-dose aspirin [18]			Lamotrigine for epilepsy [19]		
	Effect	95 per cent CI	Test of homogeneity	Effect	95 per cent CI	Test of homogeneity	Effect	95 per cent CI	Test of homogeneity	Effect	95 per cent CI	Test of homogeneity
OR												
Fixed effect	0.07	0.03 to 0.17	$Q=4.15$ d.f. $=6$	1.31	1.03 to 1.68	$Q=10.8$ d.f. $=4$	3.88	3.35 to 4.49	$Q=67.5$ d.f. $=63$	2.87	1.92, 4.29	$Q=5.71$ d.f. $=9$
Random effects	0.09	0.04 to 0.24	$P=0.66$	1.38	0.90 to 2.11	$P=0.03$	3.60	3.07 to 4.22	$P=0.33$	2.83	1.89, 4.25	$P=0.97$
RD												
Fixed effect	−0.0003	−0.0002 to 0.0004	$Q=104.9$ d.f. $=5$	0.05	0.01 to 0.09	$Q=8.3$ d.f. $=4$	0.23	0.21 to 0.25	$Q=85.8$ d.f. $=63$	0.15	0.10, 0.20	$Q=14.72$ d.f. $=9$
Random effects	−0.0004	−0.0002 to −0.0007	$P<0.001$	0.06	−0.01 to 0.12	$P=0.08$	0.22	0.20 to 0.25	$P=0.03$	0.12	0.06, 0.18	$P=0.10$
RR(B)												
Fixed effect	1.0003	1.0003 to 1.0004	$Q=110.3$ d.f. $=6$	1.21	1.02 to 1.43	$Q=12.7$ d.f. $=4$	2.47	2.23 to 2.73	$Q=91.8$ d.f. $=63$	2.32	1.67, 3.23	$Q=4.62$ d.f. $=9$
Random effects	1.0005	1.0002 to 1.0007	$P<0.001$	1.28	0.92 to 1.77	$P=0.01$	2.29	2.01 to 2.61	$P=0.01$	2.28	1.63, 3.18	$P=0.87$
RR(H)												
Fixed effect	0.07	0.03 to 0.16	$Q=4.17$ d.f. $=6$	0.93	0.88 to 0.99	$Q=6.4$ d.f. $=4$	0.73	0.70 to 0.75	$Q=112.8$ d.f. $=63$	0.83	0.78, 0.89	$Q=17.24$ d.f. $=9$
Random effects	0.09	0.04 to 0.24	$P=0.65$	0.92	0.85 to 0.99	$P=0.18$	0.75	0.72 to 0.79	$P<0.001$	0.88	0.81, 0.95	$P=0.05$

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analyses show significant benefit of this treatment, but the summary statistics are significantly heterogeneous on all but the odds ratio scale (Table I).

3.4. Lamotrigine add-on therapy for treatment of epilepsy [19]

The fourth review pools results from ten placebo-controlled trials of lamotrigine add-on therapy for drug-resistant partial epilepsy in patients of all ages. The outcome is a 50 per cent reduction in seizure frequency. Nine of the ten trials were undertaken in adults with an average placebo response rate of 8.1 per cent. The tenth trial recruited children, and had a higher placebo response rate of 15.8 per cent. All analyses showed statistically significant benefit of the treatment, only the risk ratio analysis of no response showing statistically significant heterogeneity (Table I).

4. PATTERNS OF ABSOLUTE BENEFIT ASSOCIATED WITH CHOICE OF SUMMARY STATISTIC

4.1. Graphical representations

The four summary statistics, the odds ratio, the risk difference, and risk ratios of beneficial and harmful outcomes correspond to different patterns of the relationship between event rates in the control and experimental groups. The shapes of these patterns can be depicted on L'Abbé plots, in which the event rate in the treatment group is plotted against the event rate in the control group. Figure 1 depicts contours of constant treatment effect for each possible measure.

Jimenez *et al.* proposed that plotting trial results on a L'Abbé plot may shed light on whether a chosen effect measure is likely to be a good overall summary, according to whether or not the points follow a particular shape [20]. In practice this is difficult to achieve, as the differences between the patterns are subtle and cannot be discerned in the presence of random error.

By plotting instead the absolute benefit of treatment against baseline risk (which is estimated from the control group event rate) the patterns can be distinguished more clearly, as illustrated in Figures 2 and 3. It is important to note that control group event rates are not being used in this approach as a predictor of the summary statistic, a process noted to be prone to bias, as discussed further in Section 7.

It is important to clearly distinguish between two clinical situations when constructing these plots, and in the empirical investigations that follow. First, the intention of some interventions is to prevent the occurrence of adverse outcomes (such as recurrence, disease progression or death), a desirable effect being a decrease in overall event rates. Such interventions are referred to in this paper as *preventive interventions.* This term is used in a broad sense to include both interventions to prevent healthy people becoming sick, and interventions to prevent sick people becoming sicker. Contours of constant treatment effects for preventive interventions are plotted in Figure 2, where baseline risk (p_{BR}) is the chance of the event without treatment, and absolute benefit the decrease in event rates due to treatment calculated as in equations (1a–1d).

For preventive interventions:

$$\text{absolute benefit} = \frac{p_{BR}(1 - p_{BR})(1 - OR)}{1 - p_{BR}(1 - OR)} \tag{1a}$$

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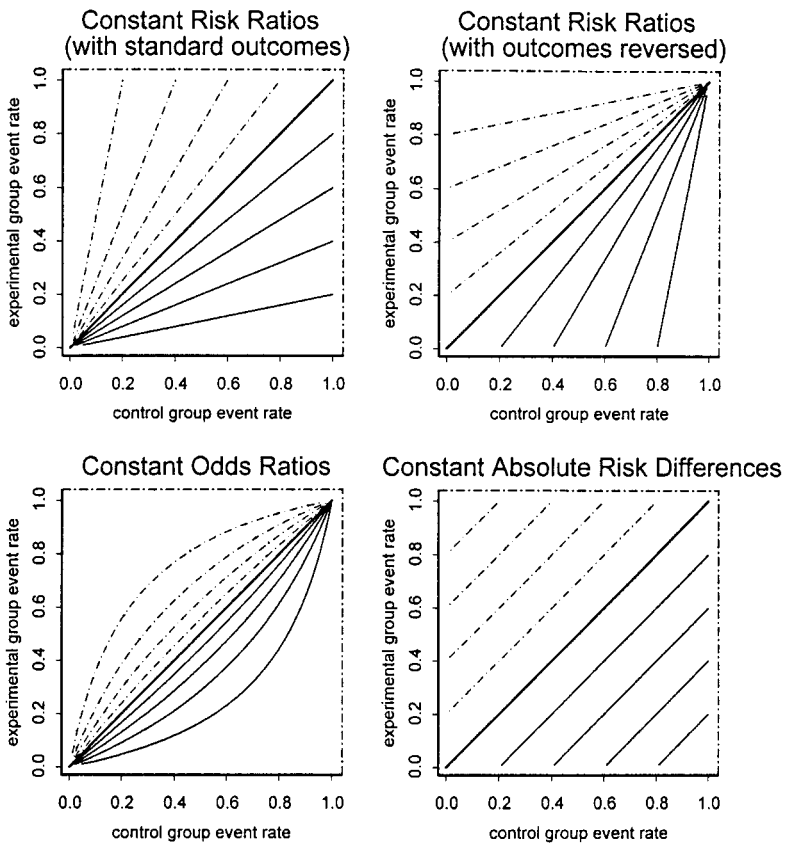


Figure 1. L' Abbé plots demonstrating constant OR, RD, RR(B) and RR(H). Lines are drawn for RR and OR of $0.2, 0.4, 0.6, 0.8, 1, 1.25, 1.67, 2.5$ and 5, and for RD of -0.8 to $+0.8$ in steps of 0.2. The bold solid line marks the line of no treatment effect ($\mathrm{RR}=1$, $\mathrm{OR}=1$, $\mathrm{RD}=0$). The solid lines indicate treatments where the event rate is reduced (or the alternative outcome is increased). The dashed lines indicate interventions where the event rate is increased (or the alternative outcome is decreased). In each case, the further the lines are from the diagonal line of no effect, the stronger is the treatment effect.

$$\text{absolute benefit} = -\mathrm{RD} \tag{1b}$$

$$\text{absolute benefit} = p_{\mathrm{BR}}(1 - \mathrm{RR(H)}) \tag{1c}$$

$$\text{absolute benefit} = (1 - p_{\mathrm{BR}})(\mathrm{RR(B)} - 1) \tag{1d}$$

Other interventions aim to induce desirable events (such as relief, recovery or cure), a desirable effect being an increase in overall event rates. These interventions are referred to in this paper as *therapeutic interventions.* Contours of constant treatment effects are shown in a similar graphic in Figure 3, but with the baseline risk (p_{BR}) being the chance of the desirable event occurring without treatment (in placebo controlled trials this is estimated by the placebo response rate) and the absolute benefit as the increase in event rates due to treatment calculated as in equations (2a–d).

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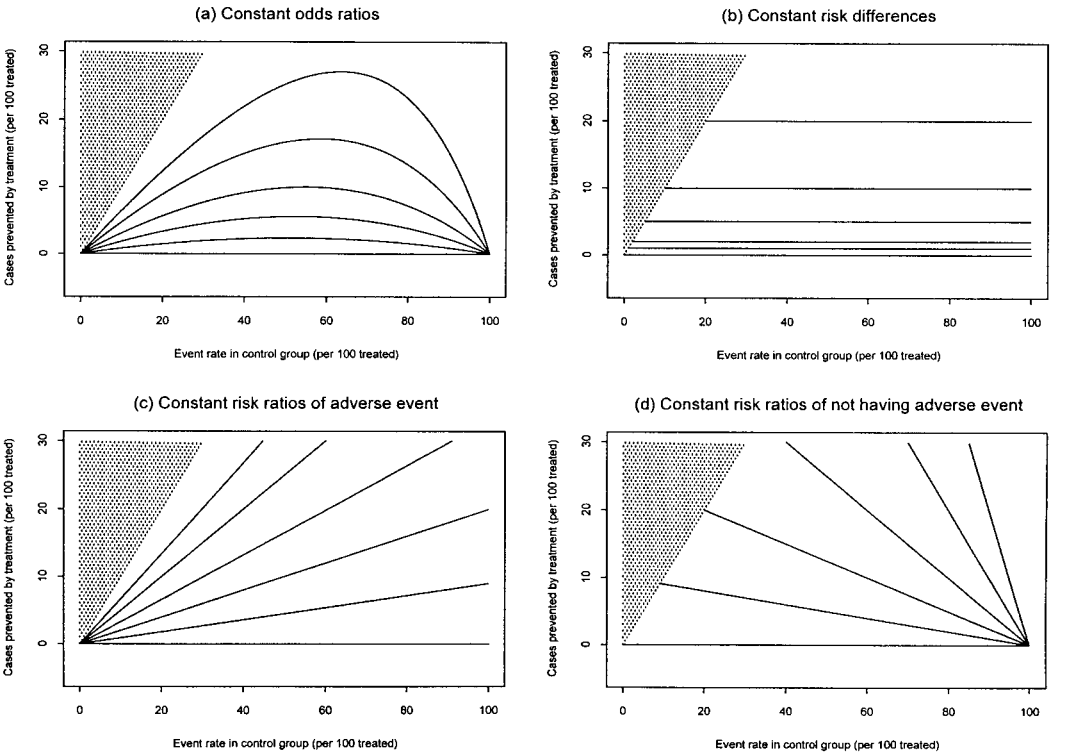


Figure 2. Patterns of predicted decreases in event rates associated with preventive interventions (expressed as the number of events prevented per 100 treated) with baseline risk (estimated by the control group event rate) for constant (*a*) OR, (*b*) RD, (*c*) RR(H) and (*d*) RR(B). Trial results cannot fall within the shaded areas of the plots. Lines are plotted for: (*a*) OR of $1, 0.91, 0.8, 0.67, 0.5$ and 0.33; (*b*) RD of $0, -0.01, -0.02, -0.05, -0.1$ and -0.2; (*c*) RR(H) of $1, 0.91, 0.8, 0.67, 0.5$ and 0.33; (*d*) RR(B) of $1, 1.1, 1.25, 1.5, 2$ and 3. In each case, the further the lines are from the horizontal of no effect, the stronger is the treatment effect.

For therapeutic interventions:

$$\text{absolute benefit} = \frac{p_{\text{BR}}(1 - p_{\text{BR}})(\text{OR} - 1)}{p_{\text{BR}}(\text{OR} - 1) - 1} \tag{2a}$$

$$\text{absolute benefit} = \text{RD} \tag{2b}$$

$$\text{absolute benefit} = p_{\text{BR}}(\text{RR(B)} - 1) \tag{2c}$$

$$\text{absolute benefit} = (1 - p_{\text{BR}})(1 - \text{RR(H)}) \tag{2d}$$

The patterns observed in the second scenario are in fact reflections of those from the first. The shading in both plots indicates areas within which trial results are impossible.

In these figures the characteristic patterns can be clearly identified as rising diagonals and falling diagonals (RRs), horizontal lines (RD) and a curved line (OR). When there is no

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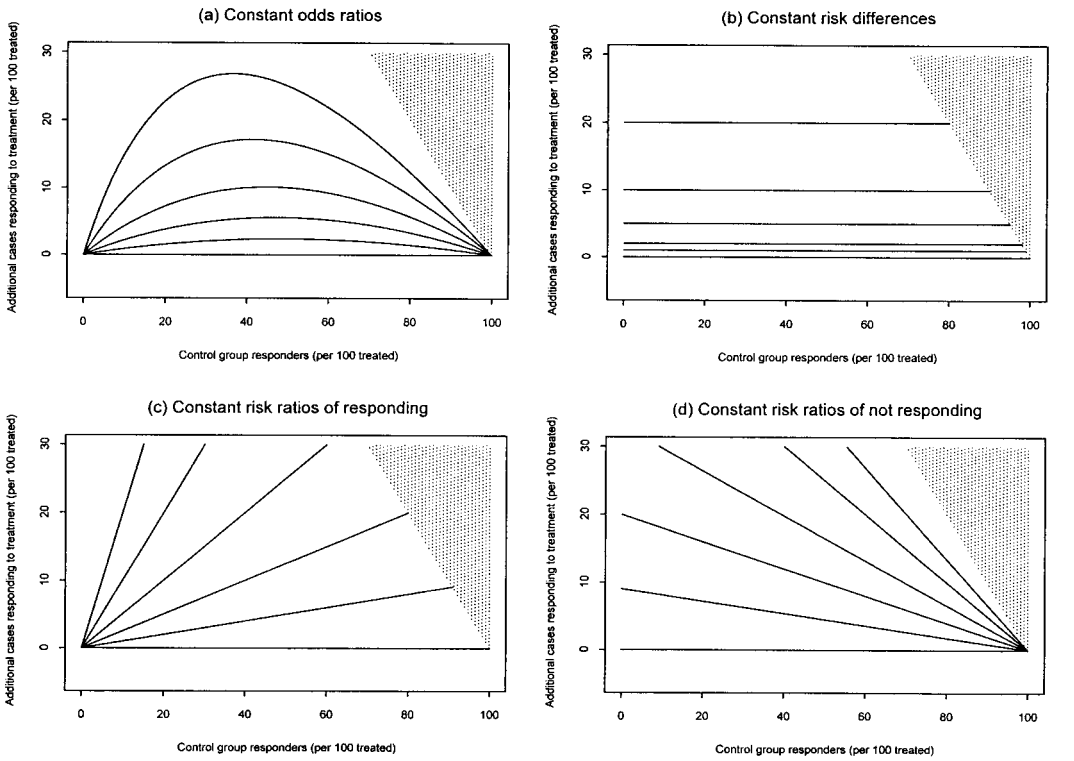


Figure 3. Patterns of predicted increases in response rates associated with therapeutic interventions (expressed as the number of extra events per 100 treated) with baseline risk (estimated by the control group response rate) for models of constant (*a*) OR, (*b*) RD (*c*) RR(B) and (*d*) RR(H). Trial results cannot fall within the shaded areas of the plots. Lines are plotted for: (*a*) OR of $1, 1.1, 1.25, 1.5, 2$ and 3; (*b*) RD of $0, 0.01, 0.02, 0.05, 0.1$ and 0.2; (*c*) RR(B) of 1,1.1, 1.25,1.5,2 and 0.3; (*d*) RR(H) of $1, 0.91, 0.8, 0.67, 0.5$ and 0.33. In each case, the further the lines are from the horizontal of no effect, the stronger is the treatment effect.

treatment effect all four models reduce to a horizontal line of zero absolute benefit. By differentiating equation (1a) with respect to baseline risk and equating the differential to zero, it can be shown that the largest absolute benefit for preventive interventions occurs in the OR model when

$$p_{\mathrm{BR}} = \frac{1}{\sqrt{\mathrm{OR}} + 1}$$

The absolute benefit of treatment (sometimes referred to as *therapeutic benefit*) has been noted as being preferred by clinicians for the application of results of research [21, 22]. Treatment decisions involve balancing benefits of treatment against potential harms for individual patients – which can only be achieved by expressing treatment effects in absolute terms. If the control group event rate is interpreted as an estimate of patients' baseline risk, Figures 2 and 3 can be used to translate summary estimates from particular meta-analyses into predictions of absolute benefit, assuming the treatment effect from the meta-analysis to be directly

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applicable to the particular patient. The figures also show that the impact of the adjustment for baseline risk on predicted absolute benefit depends crucially on the choice of summary statistic, the greatest contrast in patterns of prediction being between the two risk ratios.

4.2. Limitations of the four summary statistics

Two limitations of the summary statistics are evident from the graphs: (a) constraints on the predictions; (b) bounding. The OR model is constrained to predict absolute benefits of zero both when the control group event rate is 0 per cent and when it is 100 per cent, (see equations (1a) and (1d)) which limits its appropriateness as an effect measure. There are situations when no patients are seen to recover on control yet the treatment is beneficial, or when all patients incur an adverse event without treatment, but that use of the treatment prevents outcomes in some. Use of the odds ratio for meta-analysing such data would wrongly predict no benefit in both situations.

In contrast the RR models each have only one zero constraint. For preventive scenarios the predicted absolute benefits are zero when the control group event rate is 0 per cent for RR(H) (equation (1c)) and zero when it is 100 per cent for RR(B) (equation (1d)). For therapeutic scenarios the predicted absolute benefits are zero when the control group event rate is 0 per cent for RR(B) (equation (2c)) and zero when the control group event rate is 100 per cent for RR(H) (equation (2d)).

The shaded areas of the graphs indicate the bounding of predictions. The lines of constant risk difference intersect the shaded area, indicating that they can yield impossible predictions at certain control group event rates (with predicted event rates with treatment being either below 0 per cent or above 100 per cent). For effective treatments only risk ratios of beneficial outcomes in prevention scenarios and risk ratios of harmful outcomes in therapeutic scenarios intersect these areas, the other risk ratios yielding logical predictions in all circumstances. In practice, the bounding of RR and RD measures may not matter if values of baseline risk occur only within the unbounded part of the range. For RR it is also possible to switch the event modelled by the risk ratio to avoid bounding problems, although it is unclear whether such a strategy would always yield the most appropriate fitting model. An empirical investigation of the impact of this switching strategy is reported in Section 5.2.

4.3. Theoretical models of patterns of absolute benefit with baseline risk

For preventive interventions epidemiologists have proposed that the greatest absolute benefit is obtained by treating those most at risk of the outcome [23, 24]. This corresponds to a pattern of increasing benefit with increasing control group event rates, as described by an assumption of constant RR(H) (Figure 2(*c*)). Such a model has been used in the techniques of extrapolating results of a trial or systematic review to a particular risk group (computed as either a number needed to treat or an absolute difference in risk) [25, 26]. The pattern of increasing benefit with increasing control group event rates has been observed in several systematic reviews, and in analyses within trials stratified by risk group (for examples see references [23, 24, 27]).

For therapeutic interventions the zero constraints on the odds ratio and the risk ratio of beneficial outcomes are generally not justified, as there is no reason to believe that no absolute benefit will occur in groups with no placebo response. Conversely the constraint of zero absolute benefit when the response in the control group is 100 per cent is justifiable,

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supporting the potential use of the risk ratio of benefit. Again this pattern suggests that the greatest absolute benefit could be achieved by treating groups most likely to have the poorest outcome, as described by an assumption of constant RR(H) (Figure 3(*d*)). However, unlike for preventive interventions, no empirical evidence has been published that supports this.

Other scenarios can be considered where constraining the benefits of treatment to be zero when there is no placebo response is justified – for example, if a cancer treatment that is effective only in early disease is used in patients with late stage disease, both the remission rates and the benefits of treatment will be very low. The selection of appropriate models for therapeutic interventions is therefore likely to vary with clinical context.

4.4. Graphical presentations of motivating examples

Plotting the trials from a systematic review on the graph of absolute benefit against control group event rate allows crude assessments of the suitability of the four summary statistics to be assessed. In Figure 4 the results of the trials from the four examples described in Section 3 are plotted, with summaries of OR, RD, RR(H) and RR(B) superimposed.

The efficacy of the meningitis vaccine is an example of a preventive intervention aimed at reducing the incidence of an adverse event. The trials follow the recognized pattern with groups most at risk (populations suffering an epidemic) demonstrating greatest absolute benefit (cases of meningitis avoided). As events in these trials are rare, the pattern is described equally well both by the constant odds ratio and constant risk ratio of the harmful outcome models. In this circumstance the fact that both these models are constrained to predict no absolute benefit at zero control event is logical. The heterogeneity statistics (Table I) for these two models are similar, and orders of magnitude lower than the heterogeneity statistics for the constant risk difference and constant risk ratio of remaining free from disease.

The remaining three examples are all therapeutic interventions. The trials from the second example of eradication of *Helicobacter pylori* to cure dyspepsia show a trend for decreasing absolute benefit with increasing placebo response rates. The results therefore best fit the pattern of patients with the poorest outcomes benefiting most from the intervention. The risk ratio of cure appears to be a particularly poor fit to the data, the trials at both extremes lying some distance from the fitted line. Two clinical explanations of the observed trend have been proposed. First, that high placebo response rates are observed in groups most likely to have stress-related dyspepsia (who would not benefit from eradication therapy), and second, that groups who routinely self-medicate are more likely to show placebo response and less likely to show benefit from eradication therapy. It is unclear whether these *post hoc* rationalizations of the observed trend (based on only five trials) provide adequate justification for selection of the risk ratio of remaining dyspeptic model on which to base predictions of absolute benefit. The third and fourth examples are described in detail in Section 6.

5. EMPIRICAL EVIDENCE OF CONSISTENCY

5.1. Comparison of the odds ratio, the risk ratio and the risk difference

An empirical investigation was undertaken to assess the consistency of estimates of odds ratio, risk ratio and risk difference across a large sample of meta-analyses. The 551 meta-analyses

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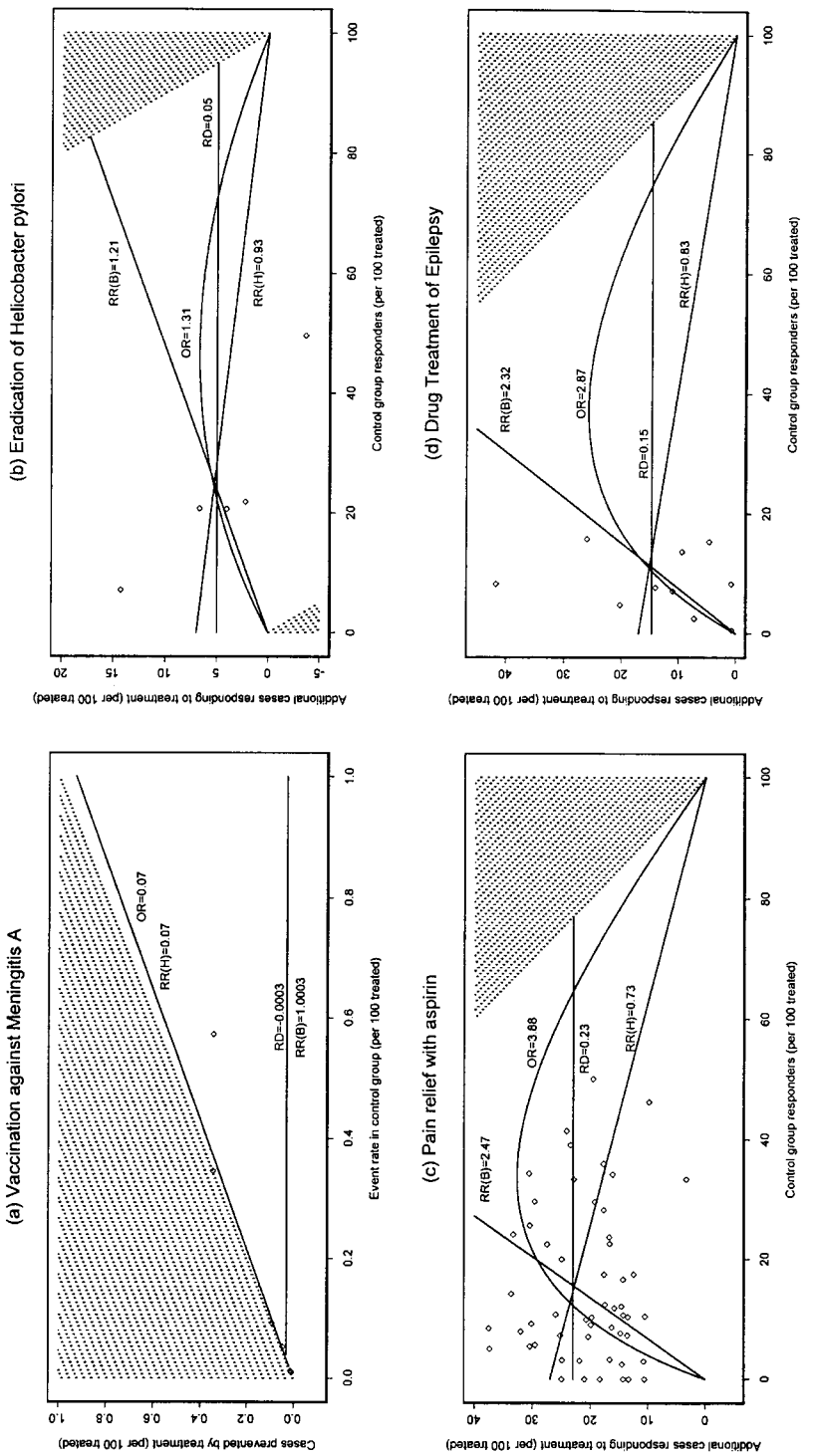


Figure 4. Plots of absolute risk difference against control group event rates for the clinical trials and meta-analytical summaries for the four examples introduced in Section 3. The lines of constant OR, RD, RR(H) and RR(B) are for the meta-analytical summaries in Table I. The shading indicates areas within which trial results cannot fall. In (*a*) only two lines appear as across the range of plotted OR and RR(H) coincide, and RD and RR(B) coincide. Labelling of axes in (*a*) differs from (*b*)–(*d*) as it is a preventive intervention rather than a therapeutic intervention.

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of binary outcomes with at least five trials published on the Cochrane Library in the Spring issue of 1997 were identified from a larger sample previously reported [28, 29]. Meta-analyses were performed using Mantel–Haenszel risk difference, risk ratio and odds ratio methods on each data set. For this analysis risk ratios were calculated using the event selected by the authors of each review. The consistency of the results for each meta-analysis was measured using the standard heterogeneity statistic, Q. The significance of the three summary statistics for each analysis were then compared.

Plots of the heterogeneity statistics for comparisons of risk ratio with the odds ratio, and of the risk ratio with the risk difference are given in Figures 5(a) and (b), respectively.

Of the 551 meta-analyses, heterogeneity was higher for risk ratio analyses than odds ratio (that is, Q for RR $>Q$ for OR) analyses in only 182 (33.0 per cent). Based on a significance level of 10 per cent, there were only 9 (1.6 per cent) meta-analyses with significant heterogeneity for RR analyses but not OR, and 13 (2.4 per cent) with significant heterogeneity for OR but not RR. The median heterogeneity statistic was lower for RR analyses (4.99) than OR analyses (5.36).

The heterogeneity of risk difference analyses was higher than risk ratio analyses (that is, Q for RD $>Q$ for RR) in 384 (69.7 per cent) meta-analyses. There were 79 (14.3 per cent) meta-analyses with statistically significant heterogeneity for RD that were homogeneous for RR, but only 10 (1.8 per cent) analyses with significant heterogeneity for RR but not RD.

Risk difference analyses were also less consistent than odds ratio analyses (that is, Q for RD $>Q$ for OR); the heterogeneity was increased in 442 (73.0 per cent). Seventy analyses (12.7 per cent) showed significant heterogeneity for RD when homogeneous for OR, only five (0.9 per cent) demonstrating significant heterogeneity for OR when not significant for RD.

Three theories concerning the relative consistency of OR and RR were investigated:

1. that there is a relationship with the average control group event rate, so that the values of the heterogeneity statistic, Q, for OR and RR analyses are similar when event rates are low, but differ considerably when event rates are high;
2. that there is a relationship with the range of control group event rates, so that where the range is high there is greater power to detect any differences that exist in heterogeneity statistics between OR and RR analyses;
3. that there is a relationship with treatment effect, with RRs greater than 1 being bounded and possibly demonstrating greater heterogeneity than ORs.

The results of these comparisons, together with comparisons with the RD, are given in Table II. Whilst there was clear evidence that increases in the average and range of control group event rates are linked to increasing heterogeneity, there was no evidence that any of these three factors is strongly related to relative consistency of RR and OR statistics; neither was there any clear relationship between the estimated overall relative risk and relative consistency of RR and OR statistics, giving no evidence that the issue of bounding impacts on the goodness-of-fit when RR >1. The RD was less consistent than both the OR and RR in all circumstances.

It therefore appears that on average risk ratio and odds ratio based analyses are equally likely to yield consistent summary statistics, and that both are likely to be more consistent than risk difference estimates. In other words, treatment effects tend to be more homogeneous across trials when expressed as relative rather than absolute effects, no obvious distinction being possible between the consistency of odds and risk ratios.

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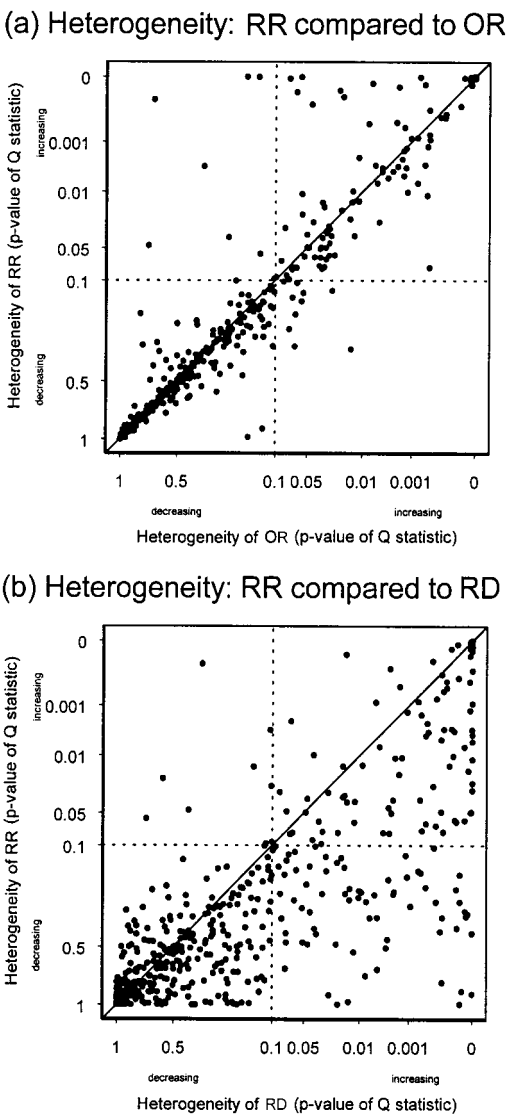


Figure 5. Pairwise comparisons of *P*-values for the heterogeneity tests of models of (*a*) constant RR against constant OR, and (*b*) constant RR against constant RD, for 551 meta-analyses from the Cochrane Library. The event chosen by the authors of each review was used in the RR analyses. Points below the line indicate meta-analyses with greater heterogeneity in (*a*) OR than RR, and (*b*) RD than RR. *P*-values are plotted on a fourth root scale [6] with the order reversed.

5.2. Investigation of the impact of switching the event for risk ratios

Most health care interventions are intended either to reduce the risk of occurrence of an adverse outcome or increase the chance of a good outcome. As already discussed, these may be seen broadly as prevention and therapeutic interventions, respectively.

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Table II. Summary of the heterogeneity observed in 551 meta-analyses from the Cochrane Library, comparing models of constant OR, constant RR and constant RD. Mean CGER is the unweighted mean of the observed control group event rates; range CGER is the difference between the highest and lowest observed control group event rates. The event chosen by the authors of each review was used in the RR analyses.

		OR	RR	RD
		Median Q (per cent where P for heterogeneity <0.1)		
All reviews	$n=551$	5.36 (22%)	4.99 (21%)	7.08 (34%)
Mean CGER 0–20%	$n=354$	4.48 (14%)	4.42 (13%)	6.15 (28%)
Mean CGER >20–40%	$n=105$	6.13 (30%)	5.81 (27%)	7.08 (36%)
Mean CGER >40–60%	$n=62$	8.21 (42%)	8.39 (44%)	11.45 (53%)
Mean CGER >60–80%	$n=19$	9.69 (42%)	11.84 (42%)	13.13 (47%)
Mean CGER >80–100%	$n=11$	16.80 (64%)	35.38 (64%)	48.04 (64%)
Range CGER 0–20%	$n=253$	3.20 (9%)	3.25 (9%)	4.29 (17%)
Range CGER >20–40%	$n=172$	6.79 (26%)	6.53 (26%)	8.91 (40%)
Range CGER >40–60%	$n=73$	9.68 (40%)	8.16 (32%)	11.76 (55%)
Range CGER >60–80%	$n=36$	12.24 (47%)	12.35 (44%)	20.61 (61%)
Range CGER >80–100%	$n=17$	22.06 (53%)	27.52 (59%)	35.50 (76%)
RR $\leqslant 0.5$	$n=68$	5.17 (28%)	4.95 (29%)	7.16 (41%)
RR >0.5–1.0	$n=283$	5.11 (22%)	4.98 (21%)	7.31 (32%)
RR >1.0–2.0	$n=173$	6.61 (21%)	5.96 (21%)	6.90 (34%)
RR >2.0	$n=27$	2.29 (11%)	2.55 (7%)	6.68 (30%)

In many situations it is natural to talk about one of the outcome states as being the *event*. In therapeutic trials participants are generally ill at the start of the trial, and the event of interest is a change of state to recovery or cure. Examples 3.2, 3.3 and 3.4 describe the use of antibiotics to eradicate *Helicobacter pylori* and cure dyspepsia, aspirin for pain relief, and lamotrigine to reduce the incidence of epileptic fits. In all situations the aim of the intervention is to improve the state of the patient. If the event is viewed as cure or improvement, the most natural and intuitive expression of relative risk is the relative probability of cure or improvement (which I have defined as RR(B)). For example, it seems more intuitive to state that 21 per cent more patients were cured of their dyspepsia with antibiotic treatment, than to state that 7 per cent fewer patients remained dyspeptic at the end of treatment (Table I). Potential disadvantages of selecting the risk ratio of a beneficial outcome are (a) that the predicted estimates will be bounded for effective treatments, especially if spontaneous cure (as described by the control group event rate) is not rare, and (b) that estimates of absolute benefit are constrained to be zero when baseline response rates are zero.

In prevention trials participants are well at the beginning of the trial and the event is the onset or exacerbation of disease, or perhaps even death. Section 3.1 describes an example of a vaccine to prevent disease. In this situation the most natural and intuitive risk ratio is the risk ratio of the adverse outcome (meningitis), which I have defined as RR(H). Vaccine efficacy is defined as the proportion of cases avoided through vaccination – the risk ratio of meningitis is 0.07 – 93 per cent of cases being avoided by vaccination. The alternative risk ratio of remaining free from meningitis of 1.0003 is both unintuitive and masks the true

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magnitude of the efficacy of the vaccine. Use of the adverse outcome also means that risk ratios are not bounded for effective treatment.

Which risk ratio of analysis is likely to be the more consistent in these two situations? Three options were considered:

1. the more intuitive outcome (implies use of RR(B) for therapeutic scenarios, RR(H) being used for preventive scenarios);
2. the greatest absolute benefit occurs in groups with the worst outcomes (implies use of RR(H) in all circumstances, which will always be unbounded);
3. the risk ratio corresponding to the least common outcome (event or non-event) will be the most consistent.

A total of 114 meta-analyses were identified in reviews first published on the Cochrane Library between 1998 and 2000, where the first outcome was binary and pooled data from five or more trials. The meta-analyses were classified according to whether the intervention was preventive or therapeutic, and outcomes classified according to whether they measured the desirable or undesirable outcome.

Each meta-analysis was analysed twice using risk ratio as a summary statistic with the events and non-events switched. Analyses were performed using the Mantel–Haenszel approach, heterogeneity statistics being calculated using Cochran's method. The significance of the heterogeneity statistics was compared to investigate whether there is evidence that analyses based on the undesirable event will more consistent than those based on the desirable outcome.

The values of the significance of the heterogeneity statistic are illustrated in Figure 6. Of the 114 analyses, 51 did not show significant heterogeneity ($p > 0.1$) on either analysis, 36 were significant on both, and 8 were more consistent for the desirable outcome whilst 19 were more consistent for the undesirable outcome (McNemar's test, $p = 0.05$).

Among the 69 meta-analyses of interventions aimed at preventing events, the risk ratio of the harmful event appeared more consistent (Figure 6(a)): of the 12 meta-analyses showing discordant heterogeneity only one was not significant for the risk ratio of the adverse outcome (McNemar's test, $p = 0.006$). There was little difference between the heterogeneity of OR and RR(H) (Figure 6(b)).

The pattern was not clear amongst the 45 therapeutic reviews (Figure 6(c)): of the 15 showing discordance 8 were significantly heterogeneous when the outcome was cure or recovery and 7 significantly heterogeneous when the outcome was remaining in the sick state (McNemar's test, $p = 1.0$). There was also no clear increase in consistency by pooling OR rather than RR(H) (Figure 6(d)).

The average control group event rates were related to interventions being preventive or therapeutic: 91 per cent of preventive interventions had adverse event rates less than 50 per cent, whilst all interventions with adverse outcome rates over 80 per cent were therapeutic interventions (Table III).

Risk ratios of the adverse outcome appeared to be more consistent than risk ratios of beneficial outcomes for all but the highest average control group event rates (Table III). For both therapeutic and preventive interventions, risk ratios of the adverse outcome were more consistent when adverse event rates were less than 50 per cent, but the patterns diverged when adverse event rates were above 50 per cent.

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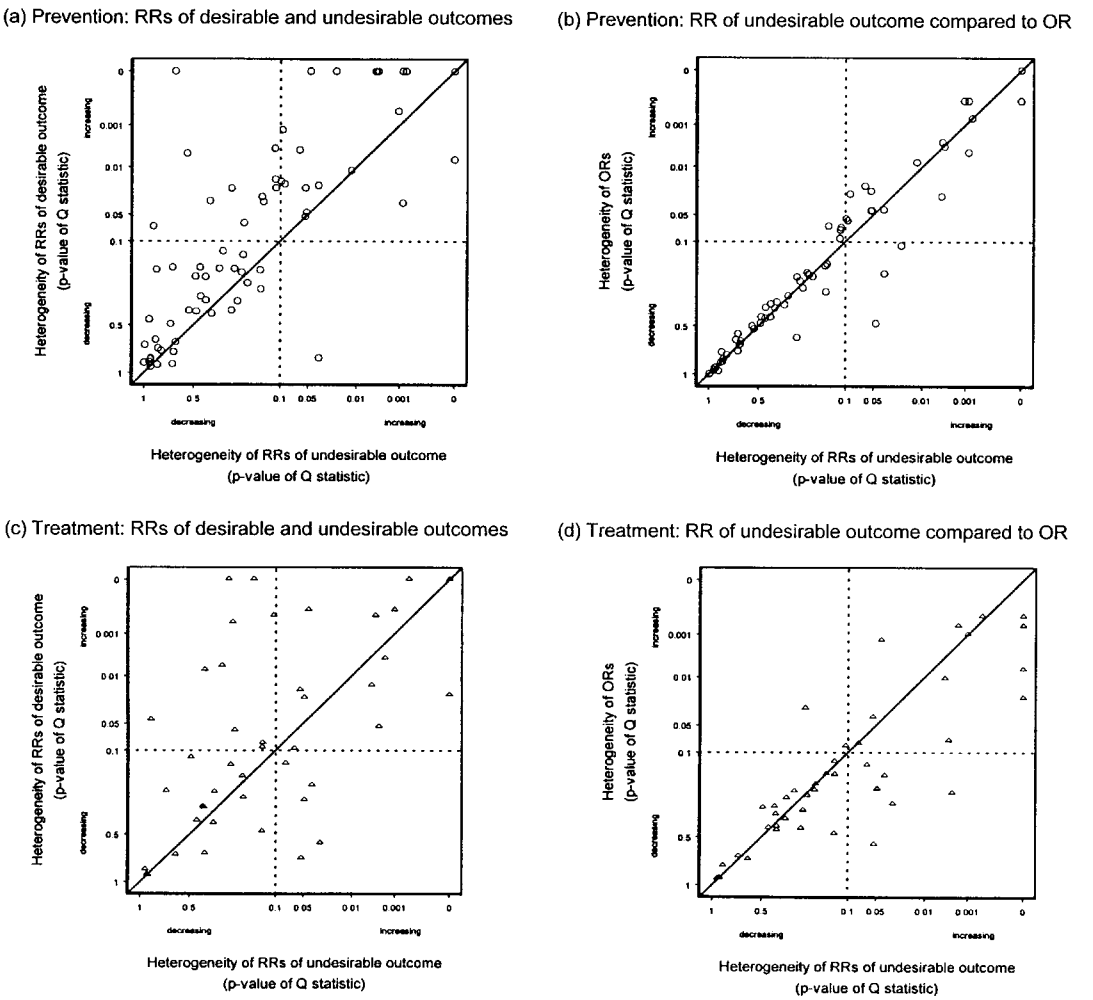


Figure 6. Pairwise comparisons of *P*-values for the heterogeneity tests of meta-analyses of (*a*) and (*c*) comparing models of constant RR(B) against constant RR(H), and (*b*) and (*d*) comparing constant OR against constant RR(H) for 114 Cochrane reviews. Graphs (*a*) and (*b*) are for 69 analyses where the event is undesirable (preventive) whilst graphs (*c*) and (*d*) are for 45 analyses where the event is desirable (therapeutic). Points above the line in all graphs indicate meta-analyses where the heterogeneity is least for the model of constant RR(H). *P*-values are plotted on a fourth root scale [6] with the order reversed.

In summary, the use of the risk ratio of the adverse outcome appears to be preferred when reviews are evaluating preventive interventions. This selection is intuitive, in agreement with the theoretical models of increasing absolute benefit with increasing risk, and avoids the problems of bounded predictions. For therapeutic interventions the selection is less clear, the only observed pattern being that estimates of risk ratios of the adverse outcome are more consistent when the adverse event rate is less than 50 per cent, although the number of reviews on which this conclusion is based is small.

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Table III. Summary of the heterogeneity observed in 114 meta-analyses from the Cochrane Library, comparing models of constant RR of the desirable (RR(B)) and undesirable (RR(H)) outcomes. Mean CGER is the unweighted mean of the observed control group event rates.

		RR(H) (Harmful outcome)	RR(B) (Beneficial outcome)
		Median Q (per cent where P for heterogeneity <0.1)	
All reviews	$n=114$	9.44 (39%)	11.49 (48%)
Prevention	$n=69$	8.15 (33%)	11.48 (48%)
Therapeutic	$n=45$	11.61 (47%)	11.36 (49%)
Mean CGER 0–20%	$n=40$	6.42 (20%)	9.77 (45%)
Mean CGER >20–40%	$n=27$	8.15 (37%)	11.15 (48%)
Mean CGER >40–60%	$n=13$	13.85 (46%)	18.80 (62%)
Mean CGER >60–80%	$n=21$	11.45 (57%)	11.69 (62%)
Mean CGER >80–100%	$n=13$	17.24 (62%)	11.01 (23%)
Prevention CGER $\leqslant$50%	$n=63$	7.50 (30%)	11.61 (46%)
Prevention CGER >50%	$n=6$	16.68 (67%)	14.30 (68%)
Therapeutic CGER $\leqslant$50%	$n=13$	8.41 (23%)	11.15 (62%)
Therapeutic CGER >50%	$n=32$	13.57 (56%)	11.53 (44%)

6. CONSIDERATION OF GOODNESS-OF-FIT

One proposed strategy for selecting a summary statistic is to choose the statistic which demonstrates the greatest degree of fit (that is, has the lowest heterogeneity statistic). As already mentioned, most meta-analyses have inadequate numbers of trials to allow formal comparison of competing models, and there are recognized problems in selecting a method of analysis based on the results of a previous statistical significance test. The third and fourth examples demonstrate two additional issues of concern.

6.1. Variation in study weights between summary statistics

The analysis of the third example of aspirin for acute pain (Section 3.3) showed significant heterogeneity in all analyses other than the OR analysis (Table I). However, the plot of absolute treatment benefit against control group event rate (Figure 4(c)) clearly shows that the odds ratio summary does not give appropriate predictions when the control group event rates are low. There are eight trials in the review with zero placebo response, all of which demonstrate substantial benefit of paracetamol – the model of constant OR predicts the benefit in these trials to be zero, as does the risk ratio of cure. The risk difference and risk ratio of remaining in pain both yield more appropriate predictions for these trials, but demonstrate significant statistical heterogeneity.

Table IV shows the weights given to the ten trials from this review (out of 64) which have the lowest control group event rates. Two sets of weights are presented: the Mantel–Haenszel percentage weights reflect the relative influence of each point in the calculation of the overall estimate, whilst the inverse variance absolute weights are used in calculation of Cochran's heterogeneity statistic, Q.

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Table IV. Relative and absolute weights, and components of the heterogeneity statistic Q for the ten trials with the lowest group event rates from example 3 (single-dose aspirin 600–650 mg for treating acute pain [18]) in meta-analyses assuming constant OR, RD, RR(H) and RR(B).

Aspirin	Placebo	Mantel–Haenszel percentage weights				Inverse variance absolute weights				Components of Q		
Cured/treated		OR	RD	RR(B)	RR(H)	OR	RD	RR(B)	RR(H)	OR	RD	RR(B)	RR(H)
6/45	0/47	0.2%	1.9%	0.1%	2.2%	0.45	379	0.47	263	0.88	3.54	1.65	7.87
9/49	0/55	0.2%	2.1%	0.1%	2.5%	0.47	325	0.48	206	1.69	0.70	2.33	2.63
17/68	0/75	0.2%	2.9%	0.1%	3.4%	0.48	365	0.49	200	3.19	0.15	6.69	0.17
8/38	0/38	0.2%	1.6%	0.1%	1.8%	0.46	229	0.48	134	1.34	0.09	2.33	0.89
4/28	0/26	0.2%	1.1%	0.1%	1.3%	0.43	221	0.46	142	0.37	1.68	1.53	3.86
5/47	0/53	0.2%	2.0%	0.1%	2.4%	0.45	473	0.47	349	0.73	7.23	1.28	14.27
14/29	0/31	0.1%	1.2%	0.1%	1.5%	0.46	120	0.50	32	3.42	7.68	1.71	3.44
14/30	0/30	0.1%	1.2%	0.1%	1.5%	0.46	125	0.50	35	3.19	6.97	1.62	3.11
7/41	1/39	1.1%	1.6%	0.3%	1.9%	1.17	286	1.33	162	0.15	2.06	1.40	3.97
4/29	1/33	1.1%	1.3%	0.2%	1.4%	1.04	235	1.21	138	0.00	3.52	1.24	5.38
										$\Sigma = 14.96$	$\Sigma = 33.62$	$\Sigma = 21.79$	$\Sigma = 45.58$

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Comparison of the Mantel–Haenszel weights between summary statistics reveals that odds ratio and risk ratio of cure models give very little relative weight to these studies with low control group event rates; risk difference and risk ratio of remaining in pain methods giving approximately 20 times more weight.

Differences in the weights used in calculation of the heterogeneity statistic (the inverse variance absolute weights) are more extreme. The absolute weights are several magnitudes larger for risk difference and risk ratio of remaining in pain than for odds ratio and risk ratio of cure. When the components of Q are calculated using these weights a paradoxical result is obtained. The two models with an appropriate prediction to these trials are deemed to be a poorer fit than the two models which predict a value completely outside the range of the points (Figure 4(c)).

Comparisons between Q-statistics generated from different summary statistics therefore do not treat the studies in the same way, which raises some concern as to whether its is reasonable to use the statistic for this purpose. Their ability to compare the relative fit of the four models is confounded by differences in the way that they weight the contributions from each study. A model may give a lower Q-statistic not because its predictions are closer to the observed results for a set of trials, but because it gives outlying trials lower weight, both when calculating the average effect and the lack of fit.

6.2. Masking sources of clinical heterogeneity

Whilst the aim of meta-analysis is to compute a single summary of the effect of an intervention, there are clinical situations where there are prior biological reasons to believe that the effectiveness of a treatment varies for reasons other than baseline risk, depending on factors such as patient characteristics. An investigation of sources of heterogeneity in a review may identify these patient groupings and provide a separate estimate of effectiveness for each. However, it is theoretically possible that important sources of heterogeneity could be missed if the strategy of using the summary with the smallest heterogeneity statistic is universally applied. It is possible that a coincidental relationship between a source of heterogeneity and baseline risk exists among the trials included in a review, such that such a source of clinical heterogeneity appears hidden when the data are analysed using one particular summary statistic.

The fourth review pools results from ten placebo-controlled trials of lamotrigine add-on therapy for drug-resistant partial epilepsy in patients of all ages. In this example only the analysis of risk ratio of no response showed statistically significant heterogeneity. A more detailed analysis of these results is given in Table V, stratifying the analysis according to whether participants were adults or children. There is no significant heterogeneity among the nine trials in adults on any effect measure. However there is a significant difference between adults and children when the treatment effect is expressed as a risk difference or risk ratio of no benefit, which could be interpreted as real heterogeneity of treatment effect. Analysis using OR or RR(B) does not detect a difference, possibly masking an increased benefit of using the treatment in children. Without clinical insight or further data it is not possible to judge which is the most appropriate analysis.

In this example it is not correct to claim that the heterogeneity disappears with a change of summary statistic, but that it becomes statistically non-significant. The magnitude of the contrasts between adults and children expressed as relative odds ratios (relative $\mathrm{OR} = 0.62$)

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Table V. Trial results and meta-analyses of OR, RD, RR(H) and RR(B) for example 4 (Lamotrigine add-on therapy for treatment of epilepsy [19]) with subgroup analyses of children and adults.

	Drug	Placebo	OR (95 per cent CI)	RD (95 per cent CI)	RR(B) (95 per cent CI)	RR(H) (95 per cent CI)
Trials in adults						
Binnie	1/16	1/18	1.13 (0.07, 19.7)	0.0007 (−0.152, 0.166)	1.13 (0.08, 16.6)	0.99 (0.84, 1.18)
Boas	6/30	4/26	1.38 (0.34, 5.53)	0.046 (−0.153, 0.245)	1.30 (0.41, 4.11)	0.95 (0.74, 1.21)
Jawad	6/12	1/12	11.0 (1.06, 114)	0.417 (0.093, 0.740)	6.00 (0.85, 42.6)	0.55 (0.30, 0.98)
Loiseau	2/11	1/14	2.89 (0.23, 36.9)	0.110 (−0.154, 0.375)	2.55 (0.26, 24.6)	0.88 (0.64, 1.21)
Matsuo	33/143	10/73	1.89 (0.87, 4.09)	0.094 (−0.011, 0.199)	1.68 (0.88, 3.22)	0.89 (0.78, 1.01)
Messenheimer	10/46	4/52	3.33 (0.97, 11.5)	0.140 (0.001, 0.280)	2.83 (0.95, 8.40)	0.85 (0.71, 1.01)
Schapel	5/20	1/21	6.67 (0.70, 63.2)	0.202 (−0.008, 0.413)	5.25 (0.67, 41.1)	0.79 (0.60, 1.03)
Schmidt	1/11	1/12	1.10 (0.06, 20.0)	0.008 (−0.223, 0.238)	1.09 (0.08, 15.4)	0.99 (0.77, 1.28)
Smith	4/41	1/40	4.22 (0.45, 39.5)	0.073 (−0.030, 0.175)	3.90 (0.46, 33.4)	0.93 (0.83, 1.04)
Overall estimate			2.46 (1.49, 4.06)	0.107 (0.052, 0.163)	2.14 (1.39, 3.29)	0.88 (0.82, 0.94)
Heterogeneity			$Q = 4.50$, d.f. $= 8$, $p = 0.81$	$Q = 7.64$, d.f. $= 8$, $p = 0.47$	$Q = 4.08$, d.f. $= 8$, $p = 0.85$	$Q = 7.22$, d.f. $= 8$, $p = 0.51$
Trials in children						
Duchowny	41/98	16/101	3.82 (1.96, 7.45)	2.60 (0.139, 0.381)	2.64 (1.59, 4.38)	0.69 (0.57, 0.83)
All trials						
Overall estimate			2.87 (1.92, 4.29)	0.147 (0.095, 0.198)	2.32 (1.67, 3.23)	0.83 (0.78, 0.89)
Heterogeneity			$Q = 5.71$, d.f. $= 9$, $p = 0.77$	$Q = 14.72$, d.f. $= 9$, $p = 0.10$	$Q = 4.62$, d.f. $= 9$, $p = 0.87$	$Q = 17.24$, d.f. $= 9$, $p = 0.05$
Adults compared to children						
Contrast of effect estimates*			0.62 (0.27, 1.44)	−0.166 (−0.298, −0.035)	0.77 (0.39,1.51)	1.31 (1.08,1.60)
Significance of difference			$P = 0.27$	$P = 0.01$	$P = 0.45$	$P = 0.007$

*Contrasts are relative effects in adults compared to children for odds and risk ratios, and difference in risk differences.

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and relative risk ratios of benefit (relative RR(B) = 1.31) whilst statistically non-significant are actually the same size or larger than the statistically significant difference in relative risk ratios of harm (relative RR(H) = 0.77) (see Table V). The difference in significance occurs due to differences in the variance and not the magnitude of the estimates.

A similar observation occurs with the one review of a preventive intervention in the empirical study in Section 5.2 that showed statistically significant heterogeneity on the risk ratio of the adverse outcome but not the beneficial outcome. One trial in this review recruited a different patient group than the other four, and used the drug at a substantially different dose – removal of this trial from the analysis removes heterogeneity from all analyses.

7. DISCUSSION

A priori selection of a summary statistic on clinical or scientific grounds undoubtedly seems preferable to a *post hoc* selection based on comparisons of analyses. This paper has set out to explore issues related to how such an *a priori* selection could be determined. For this purpose the choice of a summary statistic may best be viewed as a choice between different mathematical models of the relationship between baseline risk and absolute benefits of treatment, which can be represented graphically as in Figures 2 and 3. Significant variation in control group event rates between trials in a review must reflect diversity between trials in patient characteristics, control group interventions, definitions and assessments of outcome, study quality, or length of follow-up (event rates usually increasing with time). If the causes of variation in control group event rates between the trials can be identified, and if the shape of the relationship between these event rates and therapeutic benefit be discerned, it may be possible to choose the summary statistic which most closely fits the patterns of these relationships. However, this is rarely possible for an individual review, principally due to lack of data, multiple determinants of baseline risk and the dangers of model over-fitting.

An alternative approach investigated here is to look for evidence of general patterns across many meta-analyses. The empirical investigation presented here of data from 551 meta-analyses has shown that analyses of risk ratios are on average as likely to be as consistent as meta-analyses of odds ratios, and both are substantially more consistent than analyses of risk differences. These findings complement the comparison of OR and RD presented by Engels *et al.* [6].

The investigations have also revealed that the choice of event used for risk ratio analyses is crucial to obtaining consistent predictions. The patterns of the relationships between control and experimental group event rates differ so substantially for RR(B) and RR(H) they are best considered as separate models. Rejection of the RR as a summary statistic simply because of the asymmetry property is not justified – both models describe possible clinical scenarios and both are potentially useful. The second empirical investigation found that for interventions aiming to prevent an adverse outcome, the risk ratio of the adverse outcome was most likely to be consistent. No clear pattern emerged where the event was improvement to a better health state.

Hence empirical investigations have provided a guide for the selection of a measure for preventive interventions (use of RR(H) or OR); no general guidance is available for therapeutic interventions beyond avoiding the RD.

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What is the impact of uncertainty in the selection of a summary statistic, particularly for therapeutic interventions? The therapeutic examples 3.2–3.4 all reveal that whilst estimates of absolute benefit at the average baseline risk are similar, at all other baseline risks they vary considerably. Restraint should therefore be exercised in extrapolating to other baseline risks from these results, unless there is an understanding of the determinants of baseline risk and the impact that they have on treatment effects. Similarly, unexplained heterogeneity may indicate inadequate model fit, and similar caution exercised.

Several authors have noted the problem of investigating variation in measures of treatment effect with control group event rate [30, 31], and developed methods for correctly estimating such relationships accounting for the correlation between control group event rates and treatment effects [32–34]. Although the plots I have used depict treatment effect as a function of control group event rate (Figures 2, 3 and 4), selection of the best fitting model (OR, RR(H), RR(B) or RD) does not use regression techniques and is not troubled by the functional dependence of the treatment effect on control group event rate. The aim of the process I have discussed is to identify the statistic that is most likely to be constant across all aspects of the diversity between the trials, including control group event rates, so as to avoid the need to estimate relationships between treatment effects and baseline rates. Notably, in an investigation of 115 meta-analyses, Schmid *et al.* [35] found that relationships between baseline risk and treatment benefit are least common when the treatment effects are expressed as relative effects, significant relationships only observed for 13 per cent of meta-analyses of RR and 14 per cent of meta-analyses of OR, compared to 31 per cent of RD.

The rationale for using risk ratios over odds ratios is strong; they are a relative effect measure likely to be consistent when there are variations in event rates between trials, and have an intuitive appeal, making them accessible to users of research [2]. Also, they have been used as the basis of models describing the individualization of results from RCTs and meta-analyses to individual patients [25, 26]. The inferior mathematical properties of the RR measures do not prevent their use.

There is plenty of anecdotal evidence that the alternative relative effect measure, the odds ratio, is misinterpreted both by those reporting research studies and those using research. Where the event rates are moderately raised (as is commonly the case in randomized controlled trials), misinterpreting a treatment effect expressed as a relative reduction in the odds as a relative reduction in risk always overestimates treatment effect. In some situations the magnitude of the overestimate can be substantial [36]. This applies for both the benefits and harm associated with treatment. There are many published examples where authors have misinterpreted odds ratios from meta-analyses as if they were risk ratios [3, 36]. Indeed, Schwartz *et al*. observed that 'odds ratios are bound to be interpreted as risk ratios' [37]. A survey has indicated that significantly fewer medical practitioners have a technical understanding of the odds ratio than the risk ratio [38]. There must therefore always be some concern that routine presentation of the results of systematic reviews as odds ratios will lead to frequent overestimation of the benefits and harms of treatments when the results are applied in clinical practice.

While there are strong advocates of the odds ratio, statisticians and epidemiologists have argued that the odds ratio is often not the most suitable choice of summary statistic for summarizing the results of randomized trials and systematic reviews [39–41]. Finney has commented 'without evidence [of constancy of effect across subgroups] the average odds ratio has little

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meaning and the value for a subpopulation scarcely tells me anything I want to know... the use of the odds ratio by epidemiologists requires to be justified by epidemiological theory or empirical finding and not only by statistical convenience' [42]. The empirical investigations presented in this paper fail to indicate any evidence favouring the odds ratio over the risk ratio as a general estimator. Conversely, an additional constraint on the OR estimator has been noted: that of giving zero predictions of absolute benefit at both ends of the scale of baseline risk.

The common use of the odds ratio as the summary measure for a systematic review may have arisen for reasons of history and convenience rather than serious consideration that the odds ratio is the most appropriate model. The Mantel–Haenszel odds ratio method was published as a statistical method for the stratified analysis of case-control studies, for which the odds ratio is the only valid summary measure of association [43]. When meta-analyses of clinical trials were first undertaken in health care, the analogy between pooling trials and pooling strata was noted, and the method was reconceived as a meta-analytical method. The widespread use of the method has been supported by the availability of software to undertake the calculation, and the simplification and extension of the method by Richard Peto for pooling data from survival analyses [44]. Since then, meta-analytic methods have been developed for summarizing risk ratios and risk differences [39].

The investigations have also revealed problems in the proposed approach of selecting one statistic for analysis (most likely a relative measure) whilst presenting the results using another (most likely an absolute measure) [6]; the predictions obtained in absolute terms will always depend on the original choice of summary statistic. Global use of odds ratios for analysis and NNTs for presentation will always predict no benefit when event rates are very low or very high, with a pattern rising to a maxima at an event rate of $1/(1+\sqrt{\mathrm{OR}})$. As discussed this will not be an appropriate pattern in all circumstances. Different results would be obtained if RR(B) or RR(H) were used for the original analysis.

It was also noted that when heterogeneity statistics are computed using the standard methods, different weights are used depending on the summary statistic considered, although in all instances the statistic is considered to approximate to a chi-squared distribution with $k-1$ degrees of freedom (where k is the number of studies contributing to the meta-analysis). The impact of the use of different weights on the overall aggregated findings reported in this paper is unclear. The findings concerning the relationship between control group event rates and choice of statistic should be treated with particular caution, as they are the ones most likely to reflect changes in the weights given to outliers rather than improvements in the consistency of the findings. Alternative ways of summarizing heterogeneity should be developed which are not so dependent on the choice of summary statistic [45].

In this paper the debate concerning selection of a summary statistic for meta-analysis has moved from considering and contrasting particular properties of the competing statistics, to discussion of the statistics as four contrasting models of patterns of absolute benefit with changing control group event rates (baseline risks), and the impact that this has when meta-analytical summaries are applied to clinical practice. When choosing a summary statistic it is impossible to avoid making an assumption about the pattern of benefit related to baseline risk. I have argued that the selection of a summary statistic should not be based on a preference for superior intuitive or mathematical properties, but by thoughtful consideration of the dynamics of these models, selecting the model most likely to be a consistent estimator of treatment benefit for a particular clinical situation. Where the dynamics are

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not understood the application of the results of a meta-analysis must be made with caution. The two empirical investigations reported in this paper, and data presented elsewhere, give some guidance on which models are most likely to be consistent in particular circumstances.

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